

TYRONE EKHATOR

Software Engineer | Full-Stack Developer | Python & JavaScript Developer

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PROFESSIONAL SUMMARY

Software Engineer with 1 year of hands-on experience building full-stack web applications, backend systems, machine learning applications, database-driven platforms, and user-focused interfaces. Skilled in Python, JavaScript, Django, FastAPI, React, Next.js, Node.js, SQL, MongoDB, and machine learning workflows. Experienced in developing CRUD systems, REST APIs, authentication flows, role-based access control, admin dashboards, database models, Excel import/export features, and responsive frontend interfaces. Strong project experience across business record management, stock price prediction, real estate price prediction, event websites, and property platforms. Able to design, build, test, debug, document, and improve software applications from backend logic to frontend user experience.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, HTML, CSS

Frontend: React, Next.js, Streamlit, EJS, Tailwind CSS, Bootstrap-style UI, Recharts

Backend: Django, FastAPI, Node.js, Express.js

Databases: Microsoft SQL Server, MySQL, MongoDB, SQLite, PostgreSQL basics

Machine Learning & Data: Pandas, NumPy, Scikit-learn, TensorFlow/Keras, Random Forest, LSTM, Joblib, yfinance

Tools & Platforms: Git, GitHub, VS Code, Postman, Vercel, PowerShell

Other Skills: REST APIs, CRUD systems, authentication, role-based access control, database modelling, file uploads, Excel import/export, audit logs, responsive UI design, debugging, documentation

PROFESSIONAL EXPERIENCE

AI Software Engineer

Niger Delta Power Holding Company Limited | 1 Year Experience

- Built and improved software solutions using Python, JavaScript, web frameworks, databases, and machine learning tools.
- Developed backend features including database models, CRUD workflows, authentication-related logic, routing, validation, and file-handling features.
- Created responsive user interfaces and dashboards using React, Next.js, Streamlit, EJS, HTML, CSS, Tailwind CSS, and Bootstrap-style layouts.
- Worked with databases including Microsoft SQL Server, MySQL, and MongoDB for storing records, users, comments, property data, and application content.
- Implemented data processing and machine learning workflows using Pandas, NumPy, Scikit-learn, TensorFlow/Keras, Random Forest models, and LSTM models.
- Built reporting, charting, filtering, pagination, Excel import/export, admin workflows, and audit-log features for business-focused applications.
- Used Git and GitHub for version control, repository management, project documentation, and publishing completed software projects.
- Debugged software errors, improved UI usability, updated application flows, and documented setup/run instructions for project users.

PROJECTS

MD Registry System — Django Records Management Platform

GitHub: <https://github.com/Tyroneekhator/md-registry-system>

Built a Django-based office records management system for tracking incoming and outgoing documents, memos, invoices, attachments, approvals, and audit activity across departments.

Technologies: Python, Django, Microsoft SQL Server, HTML, CSS, JavaScript, openpyxl, ReportLab

Key Achievements:

- Developed a multi-app Django system with separate modules for accounts, records, departments, workflow requests, and audit logs.
- Built records management features including create, view, edit, filtering, sorting, pagination, attachments, and soft delete.
- Implemented role-based access control for Admin, Clerk, and Viewer users.
- Added sign-up approval and rejection workflows where Admin users can approve or reject new account requests.
- Implemented Excel import/export functionality using openpyxl.
- Added audit logging for important activities including login, logout, sign-up rejection, record actions, and workflow activity.
- Supported soft-delete and restore workflows for deleted records.
- Configured the application to support multiple users accessing the same system on an office Wi-Fi network.

Stock Price Prediction System — React + FastAPI Machine Learning App

GitHub: <https://github.com/Tyroneekhator/stock-price-prediction-react-project>

Built a full-stack stock price prediction web application that allows users to enter a stock ticker, retrieve historical stock data, run a machine learning prediction model, and compare actual closing prices with predicted prices.

Technologies: React, TypeScript, Vite, Tailwind CSS, Recharts, FastAPI, Python, TensorFlow/Keras, Scikit-learn, Pandas, NumPy, yfinance

Key Achievements:

- Developed a modern React dashboard with sidebar navigation, metric cards, stock ticker input, charts, and responsive layout.
- Built a FastAPI backend to connect frontend requests to machine learning prediction logic.
- Integrated Yahoo Finance stock data using yfinance.
- Preprocessed historical stock data using Pandas, NumPy, and MinMaxScaler.
- Used an LSTM model to generate stock closing-price predictions.
- Returned prediction results, chart data, and metrics from the backend API to the frontend.
- Displayed actual vs predicted stock prices using Recharts.
- Created multiple frontend pages including dashboard, stock directory, prediction workspace, results page, and about page.

Nigerian Real Estate Price Prediction App — Streamlit Machine Learning App

GitHub: <https://github.com/Tyroneekhator/real-estate-price-prediction-streamlit>

Built a machine learning web application that predicts Nigerian house prices based on property details such as bedrooms, bathrooms, house type, town, and state.

Technologies: Python, Streamlit, Pandas, NumPy, Scikit-learn, Random Forest Regressor, Joblib, Matplotlib, Seaborn, Geopy, Folium, Streamlit-Folium

Key Achievements:

- Developed a Streamlit application with separate pages for price prediction and real estate market insights.
- Trained a Random Forest Regressor model to estimate Nigerian house prices.
- Cleaned and prepared housing data by handling outliers, renaming columns, removing unused fields, and creating calculated features.
- Encoded categorical values such as house type, town, and state using LabelEncoder.
- Saved trained models and encoders using Joblib for reuse inside the application.
- Built a prediction interface where users can select property details and receive an estimated price in Nigerian Naira.
- Created visual insights showing average prices by state, house type, expensive towns, and price-per-room analysis.
- Integrated map display using Folium and Streamlit-Folium.

RapNation Event Website — Node.js Event Platform

GitHub: <https://github.com/Tyroneekhator/rap-event-website>

Built a full-stack event website for a hip-hop and rap festival, allowing users to view event information, artist lineups, stages, schedules, visitor advice, contact information, and community comments.

Technologies: Node.js, Express.js, EJS, MySQL, mysql2, HTML, CSS, JavaScript, dotenv

Key Achievements:

- Developed a full-stack event website using Express.js with an MVC-style structure.
- Created reusable EJS views and partials for pages, navigation, layouts, and shared UI sections.
- Designed a modern dark-themed responsive interface with cards, buttons, hero sections, and improved visual hierarchy.
- Built pages for homepage, artist lineup, stages and schedule, visitor advice, FAQ, contact, and comments.
- Integrated MySQL using mysql2 to store and retrieve user comments.
- Implemented a comments system where users can submit and view comments dynamically.
- Added environment-based database configuration using dotenv.
- Structured backend code into routes, controllers, models, middleware, and configuration folders.

Smart Customer Support Chatbot — AI Customer Support Automation Project

GitHub: <https://github.com/Tyroneekhator/smart-customer-support-chatbot>

Built a smart customer support chatbot project designed to automate customer enquiries, provide fast responses, and support a more efficient helpdesk-style user experience.

Technologies: AI chatbot workflows, Python/JavaScript project structure, customer-support automation, conversational UI concepts

Key Achievements:

- Designed and developed a customer support chatbot project focused on automating common user enquiries and improving response speed.
- Built the project around a realistic helpdesk-style use case, allowing users to interact with a chatbot instead of relying only on manual support.
- Implemented chatbot-style conversation flow to receive user questions, process input, and return relevant support responses.
- Created a structured project layout suitable for a portfolio, making the code easier to understand, maintain, and present on GitHub.
- Demonstrated knowledge of AI assistant workflows, including user message handling, response generation concepts, and conversational support design.
- Improved the project's professional value by positioning it as a practical business-support tool rather than only a basic chatbot.
- Applied software engineering principles such as clean structure, reusable logic, readable code organisation, and clear project documentation.
- Showed understanding of how chatbot systems can reduce repetitive support tasks and improve the user experience.
- Practised building user-focused software by designing the chatbot around real customer-support needs and simple support interactions.
- Strengthened portfolio evidence of AI, automation, backend logic, and chatbot-based application development.
- Added the project to GitHub to demonstrate version control, public project documentation, and practical development experience.
- Used the project to showcase problem-solving skills in building automated support features for common customer service scenarios

EDUCATION

BSc Software Engineering | Solent University Southampton | 2020 – 2023

CERTIFICATIONS

- Google Data Analytics Certificate

ADDITIONAL INFORMATION

- Strong understanding of full-stack development workflows.
- Comfortable building applications from scratch and improving existing projects.
- Experienced with debugging, testing, documentation, and project setup.
- Able to work with databases, APIs, user interfaces, and machine learning models.
- Familiar with GitHub project publishing and README documentation.

REFERENCES

Available on request.